

Easwaran Subramanian

+49 15141373425 | Berlin 10405, Germany | <https://eashwar-22.github.io>
hariharaneaswaran@gmail.com | [linkedin.com/in/easwaran-subramanian-47ba68141](https://www.linkedin.com/in/easwaran-subramanian-47ba68141) | github.com/Eashwar-22

INTRODUCTION

Data and ML engineer with 3+ years of experience building data pipelines, forecasting models and production ML systems. Skilled in Python, SQL, PySpark and cloud-based deployment, with a focus on reproducible workflows and automated monitoring. Highly fluent with modern AI coding tools and agentic workflows, using them regularly to build things without compromising code quality. Recently completed a Master's in Machine Learning at the University of Tübingen.

SKILLS

Programming: Python, SQL, Bash
Data Engineering: ETL, PySpark, Pandas, NumPy
Databases: PostgreSQL, MySQL, Relational Data Modelling
Cloud & Infrastructure: GCP, AWS - EC2/S3/Lambda/Athena/Glue, PySpark, Snowflake, dbt, Docker
Machine Learning: PyTorch, Scikit-learn, Hugging Face
Monitoring & CI/CD: GitHub Actions, git
Agentic Tools: Claude Code, Codex
Languages: English (C1), German (B1), Tamil (Native)

WORK EXPERIENCE

Graduate Data Researcher | Tübingen Institute for Medical Education, Tübingen **NOV 2023 - JUL 2024**

- Designed and maintained a data pipeline with built-in validation checks to clean and aggregate survey data from 200+ professionals, improving data integrity and eliminating a multi-hour manual process.
- Built a PHP-based internal tool that let researchers configure processing parameters and regenerate analyses without developer support.
- Redesigned relational data models and optimized SQL queries to improve retrieval performance, applying access-control best practices to sensitive medical records.

Machine Learning Engineer | Sony B.V., Stuttgart (Internship) **MAR 2023 - AUG 2023**

- Built reusable Python and PyTorch workflows for data preprocessing, model training, evaluation and hyperparameter search.
- Contributed to an automated Neural Architecture Search pipeline that evaluated 20+ model configurations, helping identify one that improved performance by ~10%.
- Wrote unit tests for preprocessing and evaluation pipelines, standardizing configuration steps to improve experiment reproducibility.

Data Analyst | Verloop.io, Bengaluru, India **MAR 2022 - SEP 2022**

- Built Python data pipelines to process, normalize and cluster thousands of chatbot conversations for downstream analytics and model-improvement workflows.
- Owned data pipeline automation for a live BI reporting dashboard, cutting the KPI reporting cycle by ~30% and ensuring stakeholders had reliable, self-service access to metrics.
- Defined and documented standardized data transformation logic for mapping unstructured inputs to categorical schemas, improving data consistency and downstream model accuracy (F1 0.78 → 0.85).

- Built and maintained SQL/Python data models to prepare and validate market, pricing and demand data across 15+ markets, supporting forecasting and business stakeholder decision-making.
- Built a demand-forecasting and pricing model using conjoint analysis, contributing to a reported 10-15% revenue improvement.
- Implemented constrained optimization (COBYLA) to generate pricing recommendations under business rules, including price ceilings and minimum sales volumes.
- Automated scenario-evaluation reporting across 100+ pricing scenarios, translating complex analytical output into clear revenue/margin/demand insights for business stakeholders.

RECENT PROJECTS

Adaptive German Audio Learning System

- Currently developing a personalized German-learning product that delivers 1 daily audio lesson through a 6-stage loop: listen, diagnose gaps, unpack, practise, relisten and transfer learning to new contexts.
- Uses a controlled LLM and voice-model (Whisper AI) pipeline for generating learner-specific passages, timestamped audio, structured grammar explanations, comprehension questions and automatically validated teaching material.
- Also designing a voice model setup that tracks speaking comprehension, dialect, grammar, spoken-word recognition and retention to personalize over 750 lessons from A1-B1 with custom scoring functions.

Visual Observability for Multi-Agent Workflows

- Built a Python observability adapter for OpenAI Agents SDK workflows, capturing model/tool calls, handoffs, retries, token usage, sources and artifacts as replayable JSON events.
- Tested dynamic agent discovery and parallel fan-out/fan-in using an eight-agent workflow; added 25+ Python and 8 JavaScript tests with GitHub Actions.
- Code: <https://github.com/Eashwar-22/agentarium>
- Demo: <https://agentarium-five.vercel.app>

Quantifying Neural Adaptability: A Reproducibility Study of Hypencoder [MASTER'S THESIS]

- Replicated the Hypencoder architecture (a neural information retrieval model), using hypernetworks to generate query-specific ranking weights for document scoring.
- Tested scaling limits by distilling knowledge from stronger cross-encoders and BM25 to identify scaling bottlenecks in pooled representations.
- Ran SLURM-orchestrated training on the MS MARCO benchmark to reach 0.727 nDCG@10 and improved zero-shot accuracy.
- Code: https://github.com/Eashwar-22/Neural_Adaptability_Reproduction_of_Hypencoder
- Walkthrough: <https://hypencoder-study.vercel.app/>

Adaptive LLM Router

- Built a FastAPI gateway that routes traffic between local (student) models and cloud (teacher) models, cutting external LLM API usage by 70% by offloading intent-classified tasks locally - a pattern similar to production systems like RouteLLM.
- Currently adding PII redaction and automated response-quality evaluation using DeepEval (LLM-as-a-judge), with a dashboard for tracking results.
- Code: https://github.com/Eashwar-22/adaptive_llm_router
- Demo: <https://frontend-two-omega-96.vercel.app/>

EDUCATION